

Pedram Abedi

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EDUCATION

University of Michigan

Ann Arbor, MI

Bachelor of Science, Honors Astronomy and Astrophysics, Interdisciplinary Physics

August 2023 - May 2026

- GPA: 3.897/4.00
- Highest Honors in Astronomy and Astrophysics; graduated with Distinction
- Senior Honors Thesis: "*LEGGOS II: A Strong Lens Model and Source-Plane Projection of the Clumpy Star-Forming Galaxy SGAS J111020.0+645950.8 at $z = 2.481$* "

University of California, Merced

Merced, CA

One year of coursework toward a Bachelor of Science in Physics

August 2022 - May 2023

- GPA: 4.00/4.00; transferred to the University of Michigan

RESEARCH EXPERIENCE

University of Michigan

Ann Arbor, MI

Undergraduate and Post-Baccalaureate Research Assistant, Strong Gravitational Lensing

May 2025 - Present

Advisor: Dr. Keren Sharon

- Led the construction and validation of a strong-lens model of the galaxy cluster SDSS J1110+6459 using JWST and HST imaging and spectroscopy. Results submitted for publication in the *Astrophysical Journal* (Abedi et al., 2026).
- Analyze lens-model outputs to measure magnifications, projected cluster mass, and the source-plane sizes and morphologies of strongly lensed galaxies, with uncertainties derived through MCMC sampling.
- Produce lens models and associated data products for the JWST LEGGOS and SLICE collaborations, enabling studies of star formation at cosmic noon, cluster dark-matter distributions and mergers, and wide-separation lensed quasars.
- Developed a Python based pipeline that automates photometric catalog selection of galaxy-cluster members for use in strong-lens models.

University of Michigan

Ann Arbor, MI

Undergraduate Research Assistant, Stellar Systems in Nearby Galaxies

January 2025- June 2026

Advisor: Dr. Eric Bell

- Investigated the HST GHOSTS survey fields to identify candidate globular clusters in the outskirts of nearby galaxies.
- Built a Python based aperture-photometry pipeline to measure the magnitudes, colors, and half-light radii of candidate stellar systems.
- Implemented masking and subtraction procedures for cosmic rays, contaminating sources, and stellar diffraction spikes.

SERVICE AND OUTREACH

University of Michigan, Department of Astronomy

Ann Arbor, MI

Curriculum Committee Undergraduate Representative

September 2025 - May 2026

- Selected to represent astronomy and astrophysics majors in the departmental curriculum committee, communicated student perspectives on course strengths, curricular gaps, and academic preparation.
- Evaluated course prerequisites and students' preparedness for upper-level core and elective major courses, which informed a revision of the curriculum in 2025.
- Advised the program on the application of AP and transfer credits toward degree requirements.
- Advocated for clearer course sequencing and more transparent procedures for exceptions, alternative pathways, and appropriate advising contacts.

University of Michigan, Student Astronomical Society

Ann Arbor, MI

Society Member

August 2025 - May 2026

- Shared resources and supported community building events for students seeking greater involvement in the astronomy community at the University and throughout Ann Arbor.
- Informally advised underclassmen navigating astronomy coursework and course selection.

COURSE RESEARCH PROJECTS

Galactic Orbits of Open Clusters — Astronomy 399, University of Michigan

Ann Arbor, MI

Fall 2024

- Developed a Python pipeline using Gaia DR3 data to model the Galactic orbits of 15 open clusters spanning young, intermediate, and old age groups.
- Integrated the orbits of young and intermediate-age clusters over 1 Gyr and old clusters over 2 Gyr to investigate their long-term evolution in the Galactic potential.
- Used Monte Carlo sampling to propagate observational uncertainties through the orbital calculations.
- Compared the orbital evolution of clusters across age groups and characterized excursions from the Galactic plane.

Physical Properties of the Coma Berenices Open Cluster — Astronomy 402, University of Michigan

Ann Arbor, MI

Fall 2025

- Identified likely members of the open cluster Melotte 111 using Gaia DR3 astrometry and photometry
- Determined the cluster's age and metallicity by fitting PARSEC isochrones spanning $\log(\text{age/yr}) = 8.6\text{--}9.0$ and $[\text{Fe}/\text{H}] = -1.0$ to $+1.0$ to the observed color-magnitude diagram.
- Used bootstrap resampling to estimate uncertainties in the fitted cluster parameters.
- Derived stellar masses, luminosities, and effective temperatures from the best-fitting model and measured separate mass-luminosity relations for the low- and high-mass cluster members.

HONORS AND AWARDS

Jerome and Isabella Karle Physical Sciences Award, Goldstein Honors Prize, LSA Honors Program

May 2026

Achievement in Research and Academics Award, LSA Department of Astronomy

May 2026

Excellence in Persian Language Award, LSA Department of Middle East Studies

May 2025

James B. Angell Scholar, University of Michigan

Fall 2025

University Honors, University of Michigan

All Terms

Dean's Honor List, University of California, Merced

Fall 2022

Chancellor's Honor List, University of California, Merced

Spring 2023

PUBLICATIONS

1. Abedi, P., Sharon, K., Hutchison, T. A., et al. (2026). "*LEGGOS II: A Strong Lens Model and Source-Plane Projection of the Clumpy Star-Forming Galaxy SGAS J111020.0+645950.8 at $z = 2.48$* ." Submitted to *The Astrophysical Journal*; arXiv:2606.20804.
2. Khullar, G., Florian, M., Bayliss, M. B., et al. (including P. Abedi) (2026). "*LEGGOS I: The JWST LEGGOS Survey—LEnsing and Galaxy Growth: Observing Substructures—Unpacks the Nature of Clumpy Star Formation and Quenching in Gravitationally Lensed Galaxies beyond Cosmic Noon*." arXiv:2606.20845.
3. Welch, B., Khullar, G., Hutchison, T. A., Sharon, K., Abedi, P., et al. (2026). "*LEGGOS: Direct Abundances of N, O, Ne, S, and Ar in Five Lensed Galaxies at Cosmic Noon*." arXiv:2609.07851.

PRESENTATIONS

1. Abedi, P., Bell, E. "*Exploring the Outskirts of M81: Detection and Classification of Companions*" University of Michigan Undergraduate Astronomy Research Symposium, Ann Arbor, MI, 2025. [Oral presentation/poster].
2. Abedi, P., Bell, E. "*An Unusual Stellar System in the Background of Messier 83*" University of Michigan Undergraduate Astronomy Research Symposium, Ann Arbor, MI, 2026. [Oral presentation/poster].
3. Abedi, P., Sharon, K. "*Magnifying Clumpy Star Formation at Cosmic Noon with JWST and Strong Lensing*" University of Michigan Undergraduate Astronomy Research Symposium, Ann Arbor, MI, 2026. [Oral presentation/poster].
4. Abedi, P., Sharon, K. "*A Cosmic Magnifying Glass Resolves Clumpy Star Formation with JWST*" University of Michigan Undergraduate Astronomy Summer Research Symposium, Ann Arbor, MI, 2026. [Oral presentation/slideshow].

SKILLS

Programming and Data Analysis:

Python (NumPy, SciPy, pandas, Matplotlib, scikit-learn, Astropy, GalPy);
Unix/Linux command line

Astronomical Software and Models:

LENSTOOL, SAOImage DS9, PARSEC stellar-evolution models and isochrones